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TCU-Net: Transformer and Convolutional Neural Network-based Advanced U-Net for Concealed Object Detection

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ABSTRACT Camouflaged object detection (COD) seeks to find concealed objects hidden in natural surroundings. COD is challenging since it has to distinguish intrinsic similarities between foreground objects and background surroundings, unlike salient object detection. Convolutional neural network (CNN)-based approaches are proposed to overcome this challenge. However, they have inherent limitations in modeling and extracting global contexts. Although Transformer-based approaches are proposed to tackle this problem, which can maintain the semantic features of input images, they have limitations in learning localized spatial features in the limited receptive field. Therefore, one of the main challenges is to conduct accurate and robust COD while maintaining global contexts without sacrificing low-level contexts. This study proposes a novel concealed object detection and segmentation method using Transformer and CNN-based advanced U-Net (TCU-Net). TCU-Net can extract globalized semantic features using the Swin Transformer-based encoder and localized spatial features using the attentive inception decoder. In particular, multi-dilated residual (MDR) blocks connecting the encoder and decoder generate refined multi-level features to improve discriminability. Finally, the attentive inception decoder generates the final camouflaged object mask by maintaining the localized spatial information. Instead of simple up-sampling of the feature map, the attentive inception decoder conducts cascaded deconvolution through inception and attention modules. A weighted hybrid loss function is used for optimizing the model, consisting of the binary cross entropy (BCE) and intersection over union (IoU) losses. We comprehensively compared the proposed TCU-Net with previous studies by analyzing different metrics based on four public datasets, such as CAMO, CHAMELEON, COD10K, and NC4K. An ablation study was also conducted to evaluate network architectures and loss functions to verify advantages of the proposed approach. Experimental analysis on public datasets proves that the proposed TCU-Net outperforms previous approaches.

INDEX TERMS Advanced U-Net, camouflaged object detection, attentive inception decoder, multi-dilated residual block, Swin Transformer

I. INTRODUCTION

Camouflage describes the behavior of particular objects or insects that hide from their predators by adapting their body's coloring with the surroundings to avoid recognition [1, 2]. Usually, camouflaged objects take the shape of similar structures or textures to the environment, which try to blend into their surroundings.

Camouflaged object detection (COD) aims to find concealed objects hidden in their surroundings. It thus has

wide applications in medical treatment (e.g., polyp segmentation [3], COVID-19 infection segmentation [4]), agriculture (e.g., pest identification [5]), and art (e.g., camouflage painting [6]). Unlike salient object detection for identifying objects with high contrast compared with foreground and background, COD is challenging since it has to distinguish intrinsic similarities between foreground objects and background surroundings [7]. Although several

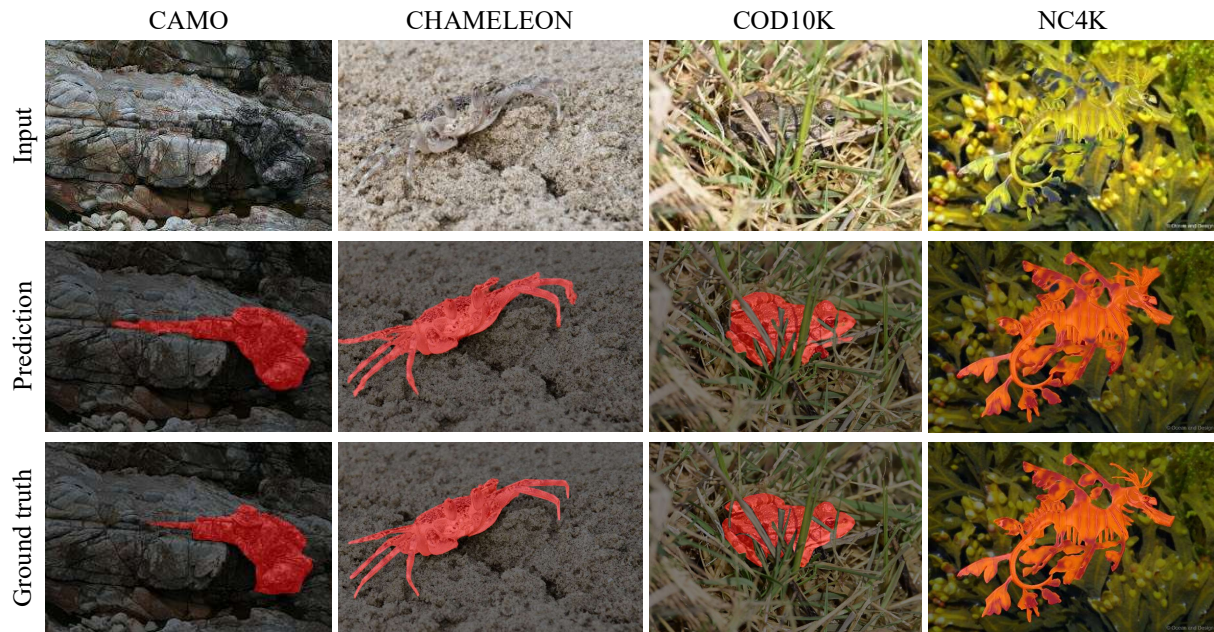


FIGURE 1. Visualization of COD results of the proposed TCU-Net.

CNN-based deep learning approaches were proposed to overcome this challenge, they still have inherent limitations in representing and learning explicit global contexts for precise and robust detection and segmentation [8-16].

Recently, the fully convolutional network (FCN) is widely used to perform semantic segmentation tasks. FCN takes an input of the varying size and generates a corresponding output for semantic segmentation with efficient inference [17]. However, it still has a generic difficulty in conducting accurate semantic segmentation since it just up-samples extracted features directly through interpolation. It is important to note that U-Net [18] also showed better performance in medical segmentation. The U-Net architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization [18]. Fan et al. [3] suggested a parallel reverse attention network (PraNet) which used a feature aggregation module instead of the decoder to integrate multi-level features obtained by each encoder block. However, these approaches still have inherent problems in COD because they have limited coverage of the receptive field due to the intrinsic locality of convolution operations [19, 20].

Recently, Dosovitskiy et al. [21] proposed the Vision Transformer (ViT) that splits an input image into a sequence of image patches and performs linear embeddings of these patches without convolutional layers for image classification tasks. It could overcome the inherent problem of CNNs mentioned above because it can extract globalized semantic features effectively. Liu et al. [22] proposed a hierarchical Transformer called Swin Transformer, which extends ViT by extracting multi-level features like CNNs. Even if Transformer-based approaches resolved limitations of CNNs, they still showed lower performance on the segmentation

tasks than CNNs because large datasets are required, and it takes much time for training [23]. Thus, it is challenging to utilize both semantic or global features and spatial or local features for effectively conducting COD.

This paper proposes the Transformer and CNN-based advanced U-Net (TCU-Net) for novel camouflaged object detection and segmentation. The proposed TCU-Net can extract both globalized semantic features using the Swin Transformer-based encoder and localized spatial features using the attentive inception decoder. The encoder and decoder can complement each other. Thus, the proposed approach can conduct COD effectively by maintaining global contexts without sacrificing local contexts. Firstly, TCU-Net extracts multi-level semantic features using the Swin Transformer-based encoder. Swin Transformer can extract multi-level context features from the entire input image, unlike CNNs that can only extract features concerning a limited receptive field. Secondly, the multi-dilated residual (MDR) block is used to enhance multi-level features extracted from Swin Transformer, which bridges the encoder and decoder. The MDR block not only enables effective feature refinement by integrating four multi-level features obtained through dilated convolutional layers but also reduces the number of feature channels and network parameters. Thirdly, TCU-Net generates a camouflaged object mask map using the cascaded CNN-based attentive inception decoder. The cascaded decoder generates the final camouflaged object map more accurately by learning the localized spatial and attention information. Thus, the decoder can complement the Transformer by effectively extracting localized features. In particular, instead of up-sampling the feature map for geometric interpolation, the attentive inception decoder conducts cascaded deconvolution

gradually through Inception and attention modules. The inception and attention modules are used to refine and learn spatial features through various convolutions and self-attention. Lastly, the deep learning network is optimized using weighted hybrid loss functions, including the weighted binary cross entropy (wBCE) and weighted intersection over union (wIoU). Fig. 1 shows the results of COD using the proposed TCU-Net, which can conduct accurate camouflaged object segmentations, almost identical to the ground truths.

We compare the proposed TCU-Net with previous studies by analyzing various metrics based on four public datasets such as CAMO[7], CHAMELEON[24], COD10K[8], and NC4K[14]. Comprehensive evaluation proves the robustness and novelty of TCU-Net. Besides, an ablation study is also performed to evaluate the proposed approach concerning network architectures and loss functions.

The main contributions of the proposed approach are described as follows:

- 1) TCU-Net has a new hybrid deep learning network composed of the Swin Transformer-based encoder that extracts semantic features and the CNN-based cascaded attentive inception decoder that can effectively learn and refine spatial features for COD.
- 2) Through multi-level convolutional operations, the MDR block refines the semantic features extracted from the Transformer-based encoder. The MDR block's feature refinement can support an effective complement between the Transformer-based encoder and CNN-based decoder, which bridges a gap between them.
- 3) The cascaded CNN-based attentive inception decoder is proposed to generate a camouflaged object map robustly and accurately. The decoder comprises an up-sampling module that sequentially increases the resolution of encoder features, an inception module that performs various convolutional operations, and an attention module that conducts refinement of inception features.
- 4) Experimental results on four public datasets verify that TCU-Net outperforms previous studies concerning various metrics, such as S-measure, weighted F-measure, mean E-measure, and mean absolute error.

The remainder of this paper is organized as follows. In Section II, the related work is reviewed and analyzed. Section III presents how the proposed TCU-Net is designed for COD. Section IV compares the experimental results of the proposed approach with previous studies and discusses the ablation study. Two seen and two unseen public datasets are used for comparative evaluation. Section V includes conclusions and future directions of this research.

II. RELATED WORK

We analyze previous studies on deep learning-based object detection and segmentation. Firstly, previous studies on salient object detection are discussed. Secondly, state-of-the-art COD approaches based on FCN or U-Net are analyzed. Lastly, Transformer-based approaches for image processing and

segmentation are reviewed since they can resolve the common locality problem of CNNs.

A. SALIENT OBJECT DETECTION

Object detection in computer vision can recognize bounding boxes or segment object regions (or masks). In particular, YOLO [25, 26] has shown excellent and effective performance in bounding box-based object detection. For this reason, some studies were conducted to detect situation elements in obscure overlapping scenes [27] and support autonomous live working robot navigation using YOLO [28]. Also, other CNN-based studies were proposed for automatic target detection in infrared imagery using dual-domain feature extraction and allocation [29] and multi-player tracking with deep identification in sport video [30].

Segmenting object regions or masks is widely used in different applications. The proposed TCU-Net focuses on object region segmentation. Wu et al. [31] proposed a cascaded partial decoder (CPD) framework for fast and accurate salient object detection. They constructed the partial decoder that discarded larger resolution features of shallower layers for acceleration, and observed that integrating features of deeper layers obtained a relatively precise saliency map. Thus, it can efficiently suppress distractors in the features and improve their representation ability. Qin et al. [32] proposed a boundary-aware salient object detection (BASNet), consisting of a densely supervised encoder-decoder network and a residual refinement module. They used a hybrid loss function that guides the network to learn the transformation between the input and ground truth in pixel-, patch-, and map-levels by fusing BCE, SSIM, and IoU losses. Zhao et al. [33] presented an edge guidance network (EGNet) for salient object detection by integrating the local edge and global location information to obtain salient edge features. The fused features can help to locate salient objects, especially their boundaries more accurately because of salient edge features. Qin et al. [34] designed two-level nested U-structure (U2-Net) for salient object detection which contains residual U-blocks without using pre-training of backbone models. Park et al. [35] proposed an advanced conditional generative adversarial network called M-GAN to conduct robust retinal vessel segmentation by repeating the encoder-decoder one more and balancing the loss function. Vania et al. [36] proposed a multistage optimization mask-RCNN (MOM-RCNN) for segmenting anatomical and functional abnormalities in the intervertebral disc. Park et al. [37] proposed an industrial surface-defect detection that segments defect areas accurately and robustly from the complex background using a deep nested convolutional network (NC-Net) with attention and guidance modules. Although these CNN-based studies showed good performance in salient object detection, they could not be directly applied to camouflaged object detection, where it is challenging to distinguish the vague boundary between foreground and background.

B. CAMOUFLAGED OBJECT DETECTION

Fan et al. [8, 16] proposed a search identification network (SINet) which includes the search module to look for the camouflaged object and the identification module to precisely detect it. They used a receptive field component [38] to incorporate more discriminative feature representations during the searching stage and extended the partial decoder component [31] with a densely connected feature. Sun et al. [9] proposed a context-aware cross-level fusion network (C2F-Net), which consists of the attention-induced cross-level fusion module to integrate the multi-level features with informative attention coefficients and the dual-branch global context module to yield multi-scale features exploiting global context information. Wang et al. [11] designed dual-branch features extraction and gradually refined cross fusion to improve COD performance by employing dense concatenation to aggregate multi-level features and expand the receptive field. Yan et al. [10] proposed a MirrorNet that leverages both instance segmentation and bio-inspired attack stream for COD by training the original and flipped image. Mei et al. [15] proposed the positioning and focus network (PFNet) that adopts the positioning module for locating the potential target objects from a global perspective and the focus module for progressively refining the coarse prediction via focusing on the ambiguous regions. Zhai et al. [12] designed a mutual graph learning (MGL) model that generalizes conventional mutual learning from regular grids to the graph domain. Lv et al. [14] presented a ranking-based COD network (Rank-Net) to localize simultaneously, segment, and rank camouflaged objects. Li et al. [13] proposed a paradigm of leveraging contradictory information to enhance the detection capability of salient object detection and COD.

C. TRANSFORMER-BASED SEGMENTATION

Zhang et al. [39] proposed TransFuse that combines a Transformer and CNN in a parallel style. It can efficiently capture global dependency and low-level spatial details in a much shallower manner using the BiFusion module that fuses the multi-level features extracted from each branch. Chen et al. [19] proposed TransUNet that merits both Transformer and U-Net, as a strong alternative for synapse multi-organ segmentation. They argue that Transformers can serve as solid encoders for medical image segmentation, combining U-Net to enhance finer details by recovering localized spatial information. Cao et al. [20] proposed Swin-UNet, a U-Net like pure Transformer for synapse multi-organ segmentation. They used a hierarchical Swin Transformer with shifted windows as the symmetric encoder-decoder architecture with skip connections to extract context features. Strudel et al. [40] proposed a Vision Transformer (ViT)-based semantic segmentation that can capture the global context effectively. Jain et al. [41] proposed SeMask that incorporates semantic information into the encoder with the help of a semantic attention operation from the pretrained hierarchical transformer-based backbones.

Recently, studies on medical polyp segmentation, transparent object segmentation, and COD were conducted to distinguish the background and background effectively. Lin et al. [42] proposed Dual Swin Transformer U-Net that concurrently incorporates the advantages of hierarchical Swin Transformer into both encoder and decoder of the U-Net architecture to enhance the semantic segmentation quality of varying medical images. Park et al. [43] proposed SwinE-Net that performs polyp segmentation by combining CNN and Vision Transformer (ViT). SwinE-Net proved accurate and effective medical segmentation by keeping global semantics Swin Transformer as well as low-level semantics of CNNs. Xie et al. [44] suggested a Transformer-based transparent object segmentation network called Trans2Seg. They used a Transformer encoder-decoder to provide the global receptive field compared with CNN's local receptive field, which shows excellent advantages over pure CNN architectures. Yang et al. [45] proposed a probabilistic representational model in combination with Transformers to explicitly reason under uncertainties, namely uncertainty-guided transformer reasoning for camouflaged object detection. They combined the benefits of Bayesian learning and Transformer-based reasoning by leveraging both deterministic and probabilistic information.

In summary, CNN-based approaches can extract spatial features in the limited receptive field but have limitations in maintaining semantic features. In particular, they have still difficulty in dealing with COD due to the vague boundary between foreground and background. Although Transformer-based studies have been proposed to handle semantic and context-aware features, they cannot robustly handle localized features due to the Transformer's generic architecture. Few studies apply a Transformer-based architecture to camouflaged object detection.

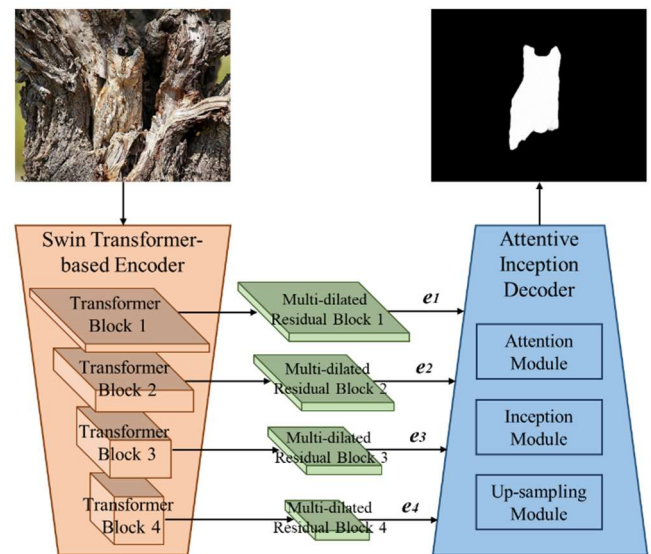


FIGURE 2. Network architecture of the proposed TCU-Net.

III. PROPOSED APPROACH

A. OVERVIEW OF THE TCU-NET ARCHITECTURE

We propose the Transformer and CNN-based advanced U-Net (TCU-Net) that can robustly and accurately conduct COD by extracting semantic, refined and spatial features globally and locally. This architecture leverages both global features from the Transformer and detailed spatial information from CNN. Transformers can extract global features through patch embedding and CNNs can extract spatial features between neighborhood pixels through convolutional filters. The multi-dilated residual (MDR) block refines the features extracted from the Transformer-based encoder and feed them to the CNN-based decoder for more accurate segmentation. The proposed TCU-Net architecture is shown in Fig. 2. TCU-Net consists of the Swin Transformer-based encoder, multi-dilated residual (MDR) blocks, and attentive inception decoder. Firstly, TCU-Net extracts semantic features using the Swin Transformer-based encoder. Secondly, the MDR block refines the semantic features by performing multi-level and multi-scale convolutional operations, which can enhance localized spatial features to improve COD performance. Lastly, the cascade attentive inception decoder generates the final camouflaged object map through up-sampling, inception and attention modules. Besides, we optimize the network using the weighted hybrid loss function, including the weighted binary cross entropy (wBCE) and the weighted intersection over union (wIoU) [46].

B. SWIN TRANSFORMER-BASED ENCODER

Semantic camouflaged object features in a given input image are effectively extracted from the Swin Transformer-based encoder [22]. The architecture of Swin Transformer is shown in Fig. 3. As a hierarchical Vision Transformer whose network is computed with shifted windows, Swin Transformer can extract features in a hierarchical format using pixel patches (windows), similar to the CNN network. Shifted windowing showed better performance since it can limit self-attention-based computation to non-overlapping local windows and allow cross-window connection. In addition, Swin Transformer showed linear computational complexity compared with ViT [21].

For an image with the input size of 384×384 , TCU-Net applies a patch partition of the patch size of 4×4 to train the first Swin transformer block. Before passing through the second Swin transformer block, patch merging is performed for the down-sampling of given features like a CNN architecture. Fig. 3(b) shows the structure of the Swin Transformer block, in which window multi-head self-attention (W_MSA) and shifted window multi-head self-attention (SW_MSA) are the core modules.

Instead of multi-head self-attention (MSA), Swin Transformer adopted window-multi-head self-attention (W_MSA) and shifted window-multi-head self-attention (SW_MSA) followed by a multi-layer perceptron with

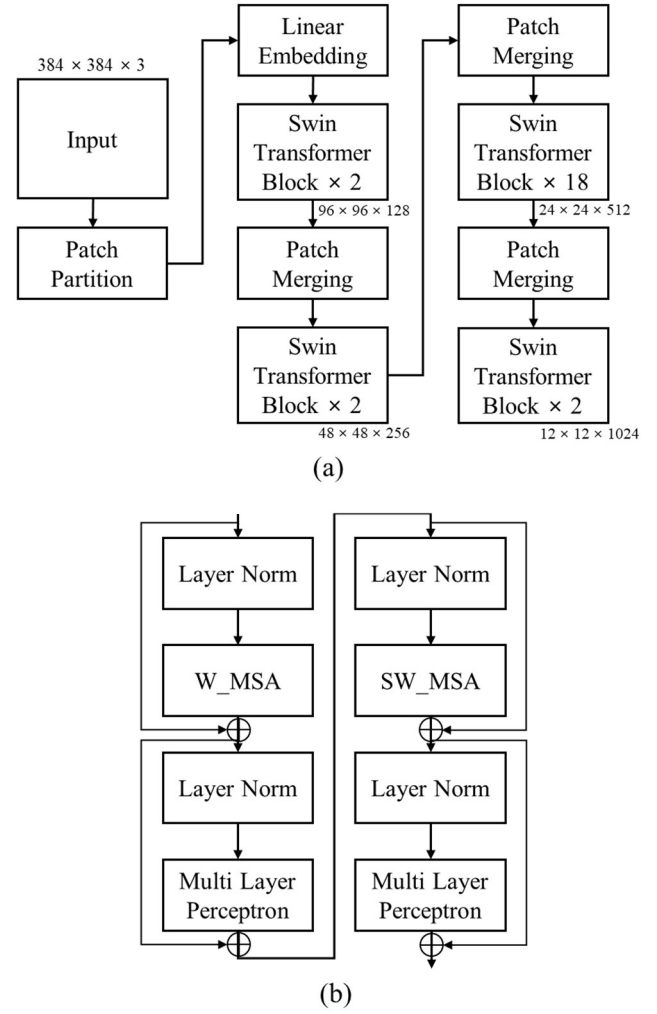


FIGURE 3. Architecture of Swin Transformer: (a) network architecture and (b) Swin Transformer block

Gaussian error linear unit nonlinearity. In addition, a LayerNorm (LN) layer follows each MSA module and MLP, and a residual connection is introduced after each module [22]. In general, since the neighboring pixels of the image are highly correlated, W_MSA performs self-attention only between patches in the window, which can reduce the amount of computation and improve detection performance. Besides, SW_MSA performs self-attention once more by shifting the window to the surrounding pixels of the patch to compensate for the disadvantage of performing self-attention only in a fixed window. The Swin Transformer module is described as follows [22]. $\hat{f}^l$ and f^l implies the output features of the W_MSA and the MLP modules of the l th block, respectively.

$$\hat{f}^l = \text{W_MSA}(\text{LN}(f^{l-1})) + f^{l-1} \quad (1)$$

$$f^l = \text{MLP}(\text{LN}(\hat{f}^l)) + \hat{f}^l \quad (2)$$

$$\hat{f}^{l+1} = \text{SW_MSA}(\text{LN}(f^l)) + f^l \quad (3)$$

$$f^{l+1} = \text{MLP}(\text{LN}(\hat{f}^{l+1})) + \hat{f}^{l+1} \quad (4)$$

Since a larger amount of training data is required to train the Transformer model compared to the CNN model [21], pre-trained weights with the ImageNet dataset [47] are used as initial weights to supplement this limitation.

C. MULTI-DILATED RESIDUAL (MDR) BLOCKS FOR FEATURE REFINEMENT

For semantic features extracted through the Swin Transformer-based encoder, the multi-dilated residual (MDR) blocks further enhance spatial information in the receptive fields with various dilations. As shown in Fig. 4, the proposed MDR block is used for feature refinement inspired by the receptive field block [38]. Each feature extracted from the Swin Transformer-based encoder is refined by passing the convolutional block for channel reduction and the other convolutional blocks with dilation = 3, 5, and 7. By using dilated convolutional layers with various sizes, different sizes of receptive fields can be efficiently learned without memory burden and loss of resolution [48]. Thus, the MDR block can minimize overfitting and vanishing gradient problems in the form of a residual that combines the results of concatenation and channel reduction of the semantic feature [49]. Besides, it can train various spatial information corresponding to multi-level and multi-scale features by reducing the last convolutional layer of each encoder feature to 64 channels. In particular, it is possible to supplement the shortcoming of the Swin Transformer that extracts multi-level features by focusing on semantic features. Therefore, it can serve as an intermediate bridge between Transformer and CNN architectures. The MDR block conducts convolutions with varying sizes of receptive fields for multi-level features, as shown in Fig. 4. Thus, it can effectively refine multi-level features [3]. Thus, TCU-Net can conduct COD more effectively by applying the MDR block in the form of feature refinement between the Transformer-based encoder and the CNN-based cascaded decoder.

Fig. 5 shows the comparative visualization of the extracted encoder features without multi-dilated residual blocks (w/o

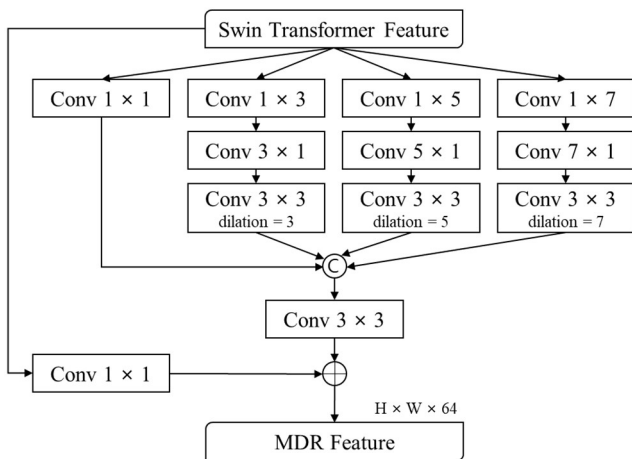


FIGURE 4. Multi-dilated residual block.

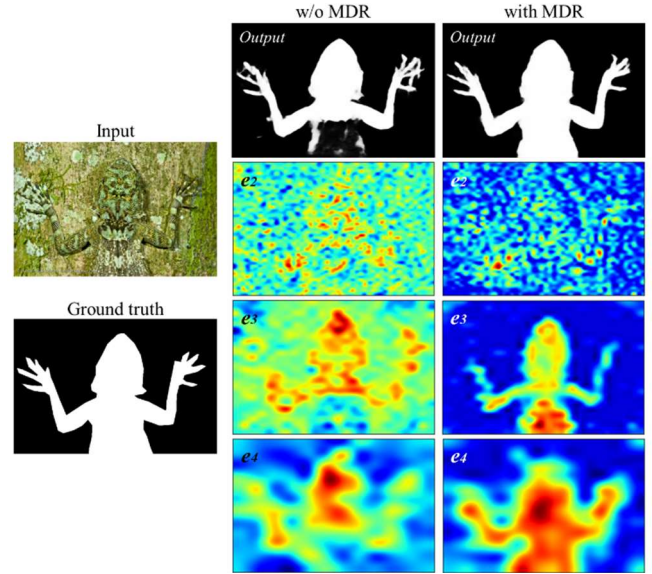


FIGURE 5. Feature visualization without and with MDR blocks.

MDR) and with multi-dilated residual blocks (TCU-Net). It presents three encoder feature maps corresponding to e2, e3, and e4 blocks in Fig. 2. For higher-level encoder features, the MDR blocks help to extract contours and semantic areas more accurately by learning spatial features than the network without using the MDR blocks. Thus, the MDR blocks play a very important role in learning low-level spatial information from the semantic features, bridging the gap between the encoder and decoder. On the other hand, the network without MDR blocks lacks in maintaining and extracting localized spatial information for higher-level features. Thus, the TCU-Net with the MDR blocks performs more accurate camouflaged object segmentation than the model without the MDR blocks.

D. CASCADED INCEPTION AND ATTENTION-BASED DECODER

The attentive inception decoder generates the final camouflaged map from the refined features that have passed through the MDR blocks, as shown in Fig. 6. The attentive inception decoder is partly inspired by the cascaded partial decoder [31]. Training proceeds sequentially from the MDR-4 feature map with the smallest resolution to the MDR-1 feature map with the largest resolution through convolutional operations, including multiplication and concatenation operations. In addition, each multi-level feature map combined through these operations passes through the inception module for learning spatial information and the attention module for learning semantic information. Finally, the final camouflaged object map is generated by applying two Inception modules and an up-sampling module to the merged MDR feature map. All the refined encoder features from low-level to high-level are fused to conduct accurate and robust COD in a cascaded way.

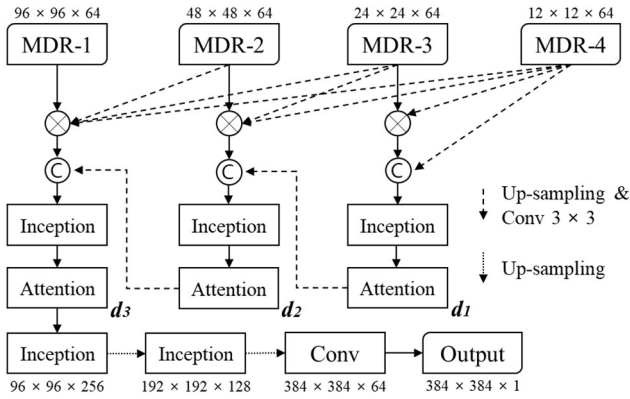


FIGURE 6. Cascaded attentive inception decoder.

Fig. 7 shows the inception and attention modules in the decoder. The inception module is designed based on the inception block of GoogleNet [50]. As shown in Fig. 7(a), it extracts spatial features through three combinations of convolutional blocks. Thus, it is possible to maintain the sharpness of the mask compared with previous studies [3, 4] that perform simple up-sampling. The attention module performs self-attention-based refinement of the decoder features that have passed the inception module, as shown in Fig. 7(b). This module sequentially infers attention maps of

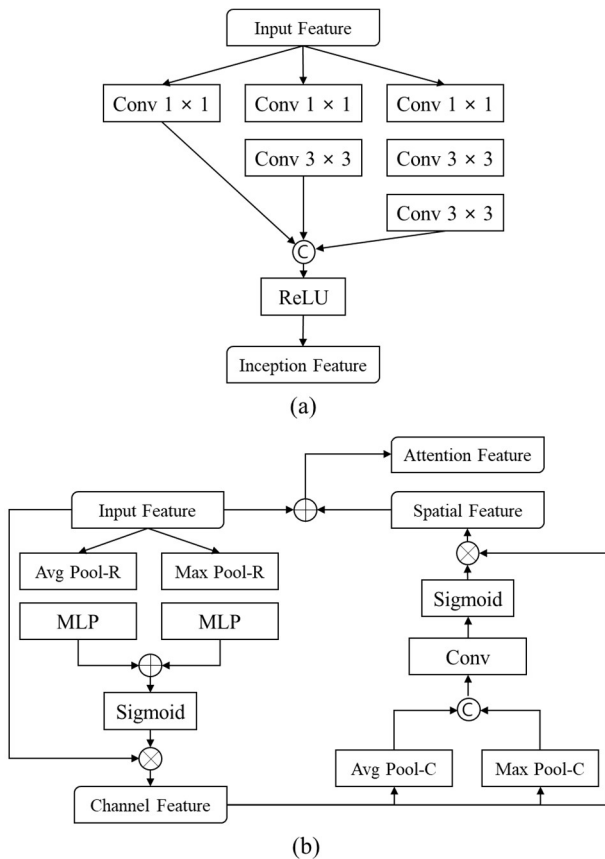


FIGURE 7. Inception and attention modules: (a) inception module and (b) attention module.

channel and spatial dimensions. The channel gate applies a multi-layer perceptron after performing average and max pooling for the resolution corresponding to the width and height of the feature, and the spatial gate applies the convolutional block after completing average and max pooling for the depth corresponding to the channel [51]. Through two gates, it is possible to learn semantic information efficiently. Thus, the attentive inception decoder can extract the refined semantic features effectively, reduce irrelevant clutter's noise, and conduct robust camouflaged object detection.

By adopting the inception and attention modules to the decoder, TUC-Net can perform feature refinement more effectively, thereby improving the performance of camouflaged object detection. By applying the inception modules, localized spatial features can be refined and globalized semantic information can be learned by the attention modules.

Fig. 8 shows the visualization result of the decoded features with the inception and attention modules and without the two modules. Three decoder features corresponding to d1, d2, and d3 blocks in Fig. 6 were visualized depending on whether or not the inception and attention modules affect the final results in the decoder. The network without the inception and attention modules generated a lot of noise and blur. However, TCU-Net generated accurate results with little noise, which guarantees robust camouflaged detection and segmentation.

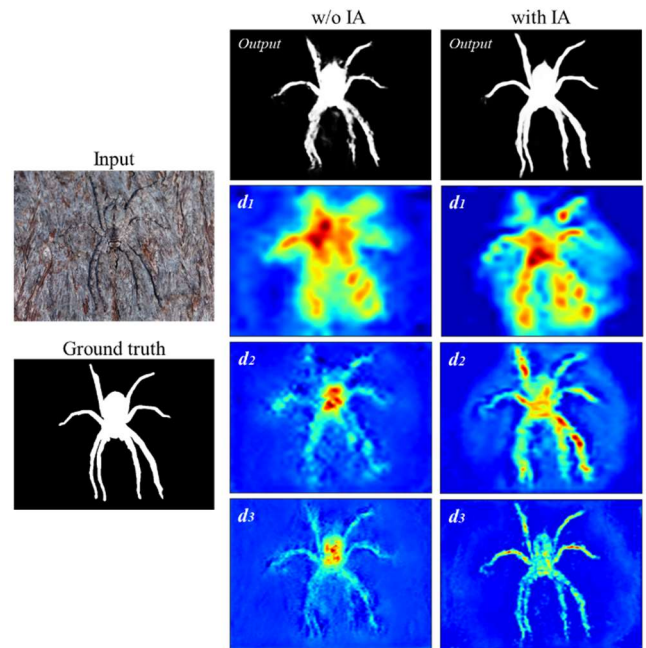


FIGURE 8. Feature visualization without and with inception and attention (IA) modules

E. LOSS FUNCTIONS

Most existing studies that support object segmentation train their networks using the binary cross-entropy (BCE) loss function. However, since the BCE loss considers only pixel-wise difference, training may be difficult in camouflaged object detection since the overall semantic context should be considered. In this study, not only pixel-wise training but also patch-wise training is supported by adopting the IoU loss and the BCE loss functions [46]. The proposed weighted hybrid loss function is defined as follows.

$$L_{\text{hyb}} = L_{\text{wbce}} + L_{\text{wiou}} \quad (5)$$

The BCE loss function computes the pixel-wise difference between the ground truth and the predicted masks. $G(x,y) \in \{0,1\}$ implies the pixel value of the ground truth (0: background, 1: defect area), where x and y are the coordinates of the pixel, and $P(x,y) \in [0,1]$ is the pixel value of the predicted mask. The IoU loss function computes the patch-wise difference instead of pixel-wise difference between the ground truth and predicted masks, which is used as an evaluation metric in the segmentation task. Both loss functions calculate $\omega(x,y)$ with different weights concerning the characteristics of the input as shown Eq. (3) and (4). γ was set to 5, and the average pooling parameters were set to the kernel size 31, stride 1, and padding 15. Thus, effective bias learning can be conducted by calculating loss weights for each input through the weighted value $\omega(x,y)$.

$$L_{\text{wbce}} = \frac{\sum_{(x,y)} [G(x,y) \log(P(x,y)) + (1 - G(x,y)) \log(1 - P(x,y))] \times \omega(x,y)}{\sum_{(x,y)} \omega(x,y)} \quad (6)$$

$$L_{\text{wiou}} = 1 - \frac{\sum_{(x,y)} P(x,y)G(x,y) \times \omega(x,y)}{\sum_{(x,y)} [P(x,y) + G(x,y) - P(x,y)G(x,y)] \times \omega(x,y)} \quad (7)$$

$$\omega(x,y) = 1 + \gamma |\text{avg_pool}(G(x,y)) - G(x,y)| \quad (8)$$

IV. EXPERIMENTAL EVALUATION

Comprehensive and comparative evaluation on COD was performed using the challenging public datasets. The experimental evaluation confirms the advantage and robustness of the proposed approach.

A. DATASETS FOR COD

Four public datasets were used for comparative evaluation of COD, such as CAMO[7], CHAMELEON[24], COD10K[8], and NC4K[14]. We trained the proposed TCU-Net with the public datasets that contain 3,040 images from COD10K and 1,000 images from CAMO. We evaluated TCU-Net with four test datasets, including CAMO, COD10K, CHAMELEON, and NC4K. Test datasets contain 250 images of CAMO, 76 images of CHAMELEON, 2,026 images of COD10K, and 4,121 images of NC4K. The training and testing assumption

is the same as the experimental setting of previous approaches [8, 16]. Each image in the dataset also includes its pixel-wise segmented ground truth. Only two seen datasets (CAMO and COD14K) were used for training, and the other two unseen datasets (CHAMELEON and NC4K) were used for evaluation purposes.

Each image was resized into an image with 384×384 for training and evaluation purpose. The reason of 384×384 resizing is to use the pretrained weights of Swin Transformer. In addition, data augmentation was conducted to generate more training datasets. Data augmentation includes rotation at 90° intervals, flipping, brightness and contrast control, salt and pepper noise generation, shearing, and square cutout, which are randomly applied to each image to increase the amount of training data.

B. IMPLEMENTATION ENVIRONMENT AND METRICS

The proposed TCU-Net was implemented in the Pytorch [52] framework, with an Nvidia RTX 3090 with the 24GB GPU memory. The batch size was set to 8, and the learning epoch to 100. The Adam optimizer is used for training the model, and the learning rate is set to $1e-5$, beta1 to 0.9, beta2 to 0.999, and epsilon to $1e-08$.

Different and well-known metrics were used for evaluation, such as structure measure (SM), mean enhanced-alignment measure (EM), weighted F-measure (FM), and mean absolute error (MAE) for the comparative evaluation. SM [53] measures the region- and object-aware structural similarity between the ground truth and predicted COD masks. EM [54] measures binary foreground map. FM [55] is another important metric for measuring both the precision and recall of the predicted mask used for balanced comparisons. FM is described as follows.

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

weighted F — measure

$$= \frac{(1 + \beta^2) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \quad (10)$$

TP, TN, and FN represent true positive, true negative, and false negative, respectively. β^2 is set to 0.3.

MAE measures the average absolute per-pixel difference between the ground truth and predicted pixels. MAE is described as follows.

$$\text{MAE} = \frac{1}{N} \sum_{(x,y)} |P(x,y) - G(x,y)| \quad (11)$$

N represents the pixel number of the input image, and $P(x,y)$, and $G(x,y)$ imply the pixel value of the predicted COD mask and that of the ground truth mask at (x,y) , respectively.

TABLE 1. PERFORMANCE COMPARISON OF THE PROPOSED TCU-NET WITH PREVIOUS APPROACHES USING SEEN DATASETS

Dataset	CAMO				COD10K			
Methods	SM↑	EM↑	FM↑	MAE↓	SM↑	EM↑	FM↑	MAE↓
BASNet [32]	0.615	0.671	0.503	0.124	0.661	0.729	0.486	0.071
F3Net [46]	0.711	0.741	0.616	0.109	0.739	0.795	0.593	0.051
CPD [31]	0.716	0.723	0.618	0.113	0.750	0.776	0.595	0.053
EGNet [33]	0.737	0.758	0.655	0.102	0.751	0.793	0.595	0.053
PraNet [3]	0.769	0.824	0.663	0.094	0.789	0.861	0.629	0.045
SINet_v1 [8]	0.745	0.804	0.702	0.092	0.776	0.864	0.679	0.043
SLSR [14]	0.793	0.826	0.725	0.085	0.793	0.868	0.685	0.041
PFNet [15]	0.782	0.852	0.695	0.085	0.800	0.868	0.660	0.040
MGL [12]	0.775	0.847	0.673	0.088	0.814	0.865	0.666	0.035
UAI [13]	0.803	0.853	0.759	0.076	0.817	0.892	0.726	0.035
SINet_v2 [16]	0.820	0.882	0.743	0.070	0.815	0.887	0.680	0.037
TCU-Net	0.882	0.933	0.844	0.042	0.880	0.941	0.804	0.020

TABLE 2. PERFORMANCE COMPARISON OF THE PROPOSED TCU-NET WITH PREVIOUS APPROACHES USING UNSEEN DATASETS

Dataset	CHAMELEON				NC4K			
Methods	SM↑	EM↑	FM↑	MAE↓	SM↑	EM↑	FM↑	MAE↓
BASNet [32]	0.847	0.883	0.795	0.044	0.698	0.761	0.613	0.094
F3Net [46]	0.848	0.894	0.770	0.047	0.782	0.825	0.706	0.069
CPD [31]	0.857	0.874	0.771	0.048	0.790	0.810	0.708	0.071
EGNet [33]	0.856	0.883	0.766	0.049	0.796	0.830	0.718	0.067
PraNet [3]	0.860	0.907	0.763	0.044	0.822	0.876	0.724	0.059
SINet_v1 [8]	0.872	0.936	0.827	0.034	0.810	0.873	0.772	0.057
SLSR [14]	0.893	0.938	0.839	0.033	0.839	0.883	0.779	0.053
PFNet [15]	0.882	0.942	0.810	0.033	0.829	0.888	0.745	0.053
MGL [12]	0.893	0.923	0.813	0.030	0.833	0.867	0.739	0.053
UAI [13]	0.894	0.943	0.848	0.030	0.842	0.898	0.771	0.047
SINet_v2 [16]	0.888	0.942	0.816	0.030	0.847	0.903	0.770	0.048
TCU-Net	0.910	0.959	0.862	0.021	0.896	0.939	0.852	0.030

C. COMPARISON WITH PREVIOUS APPROACHES

When compared with SINet_v2 [16], which showed the best performance among previous studies, the FM value of the proposed method was increased by 13.6% in CAMO and 18.2% in COD10K of the seen dataset, and by 5.6% in CHAMELEON and by 10.6% in NC4K of the unseen dataset, as shown in Table 1 and Table 2. It is confirmed that the proposed TCU-Net is superior to the state-of-the-art studies by a large margin. In addition, the fusion of the Transformer-based encoder and cascaded inception and attention-based decoder shows robust performance for camouflaged object detection. Compared with the existing methods, the proposed method simultaneously utilizes the Transformer-based encoder that effectively extracts semantic features, MDR blocks that refine the encoded features, and the cascaded

inception and attention that accurately conducts COD. Therefore, it can derive the best performance for camouflaged object detection.

The results of the qualitative visual comparison are shown in Fig. 9. Similar to the evaluation result in Table 1 and Table 2, the proposed TCU-Net proved better qualitative performance compared with previous studies. COD results for CAMO, CHAMELEON, COD10K, and NC4K datasets are shown sequentially from the first row to the fourth row. In the comparative evaluation of the first rows of CAMO and COD10K, the existing methods could not find the hidden object properly, and thus ambiguous segmentation of background (e.g., green leaves and coral reefs) or generation of a lot of noise caused false positive errors. On the other hand, the proposed TCU-Net accurately segmented the camouflaged object with little noise. For the first rows of CHAMELEON and NC4K, the existing methods did not recognize the

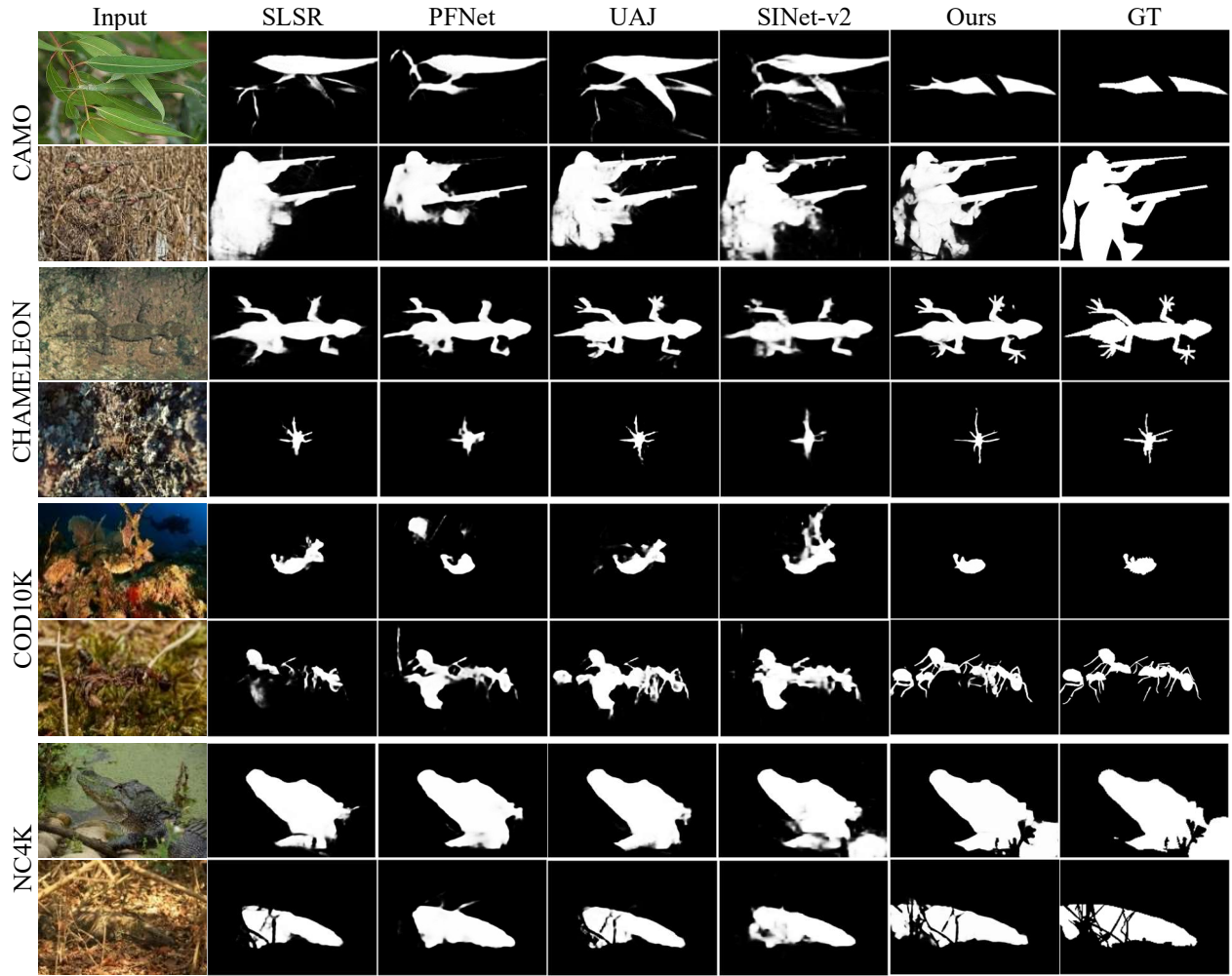


FIGURE 9. Qualitative comparison of the proposed TCU-Net with previous approaches.

boundary between the background and the camouflaged object exactly, causing a lot of blur or false negative errors in recognizing the foreground as the background. For the second rows of CHAMELEON and COD10K, the proposed approach can detect fine details of concealed objects such as ant and spider legs. The proposed TCU-Net generated less blur and maintained the sharpness of segmented objects. Concerning comparative evaluation, the proposed TCU-Net can extract camouflaged objects more concisely and robustly, unlike existing methods, by reducing both false negative and false positive errors.

D. ABLATION STUDIES

Ablation studies were conducted to confirm the advantage and novelty of the proposed TCU-Net concerning loss functions and network architectures. Loss functions are comparatively evaluated based on three different combinations: 1) BCE, 2) BCE and IoU, 3) weighted BCE and weighted IoU losses. The network architecture is further analyzed with respect to the encoder and decoder. First, the encoder is evaluated based on 1) CNN-based encoders such as ResNet and EfficientNet and

2) the proposed Swin Transformer-based encoder. Then, the decoder is evaluated based on 1) U-Net based decoder composed of transposed convolutional layers, 2) U-Net decoder with MDR blocks, 3) the proposed decoder without MDR blocks, 4) the proposed decoder with MDR blocks and without inception and attention modules, and 5) the proposed cascaded inception and attention-based decoder.

Four evaluation metrics were measured for the seen dataset (CAMO) and the unseen dataset (CHAMELEON), as shown in Table 3. The experimental results verify the advantage of the proposed loss function and network structure in the encoder and decoder. Regarding the loss function, the BCE-only loss showed the worst performance. On the other hand, the combination of wBCE and wIoU losses yielded the best performance.

Widely used encoders with CSPDarkNet (YOLOv4), ResNet and EfficientNet are compared with the Swin Transformer-based encoder. The Transformer-based encoder yielded better performance than various CNN-based encoders by a large margin. TCU-Net utilizes the Transformer-based

TABLE 3. PERFORMANCE COMPARISON OF THE ABLATION STUDIES

Dataset	Category	Configuration	SM↑	EM↑	FM↑	MAE↓
CAMO (seen)	Loss Function	BCE	0.869	0.913	0.821	0.047
		BCE+IoU	0.879	0.930	0.836	0.043
		wBCE+wIoU	0.882	0.933	0.844	0.042
	Encoder	CSPDarkNet (YOLOv4)	0.785	0.833	0.686	0.086
		ResNet	0.800	0.852	0.711	0.082
		EfficientNet	0.821	0.870	0.729	0.073
		Swin Transformer-tiny	0.773	0.841	0.669	0.085
		Swin Transformer	0.882	0.933	0.844	0.042
	Decoder	CSPDarkNet+FCN	0.767	0.805	0.646	0.095
		U-Net	0.867	0.912	0.816	0.049
		MDR+U-Net	0.880	0.927	0.831	0.044
		Without MDR	0.872	0.924	0.828	0.045
		Without Inception and attention modules	0.882	0.929	0.833	0.045
		Cascaded Inception	0.882	0.933	0.844	0.042
CHAMELEON (unseen)	Loss Function	BCE	0.886	0.919	0.815	0.027
		BCE+IoU	0.902	0.949	0.856	0.023
		wBCE+wIoU	0.910	0.959	0.862	0.021
	Encoder	CSPDarkNet (YOLOv4)	0.872	0.923	0.782	0.036
		ResNet	0.868	0.926	0.778	0.037
		EfficientNet	0.957	0.922	0.756	0.039
		Swin Transformer-tiny	0.847	0.913	0.750	0.041
		Swin Transformer	0.910	0.959	0.862	0.021
	Decoder	CSPDarkNet+FCN	0.854	0.907	0.742	0.044
		U-Net	0.901	0.935	0.835	0.024
		MDR+U-Net	0.904	0.945	0.842	0.024
		Without MDR	0.909	0.938	0.854	0.021
		Without Inception and attention modules	0.904	0.946	0.842	0.024
		Cascaded Inception	0.910	0.959	0.862	0.021

encoder capable of learning globalized semantic features for robust results.

The lightweight network and reducing calculation time are important parts in the field of deep learning. Thus, we have evaluated the Swin Transformer-based tiny model, as shown in Table 3. The tiny model has one third the size of TCU-Net. For this reason, it showed less performance than TCU-Net. On the other hand, the tiny model runs 52.08 frames per second (FPS, 0.0912 second per image) and the proposed TCU-Net runs 37.45 FPS (0.0267 second per image). There is trade-off between the model accuracy and calculation speed. It is important to note that both showed real-time performance. For future search work, it would be valuable to balance the relationship between these two factors.

Furthermore, the decoder characteristics are evaluated. We analyzed the performance of the model with CSPDarkNet+FCN architecture. Although YOLO is well known for excellent bounding box-based object detection, the combination of the encoder of YOLOv4 and the decoder of

FCN showed worse performance than TCU-Net. Since the U-Net decoder performs up-sampling using only simple transposed convolutional operation, it performs worse than the proposed TCU-Net. Since the proposed cascaded decoder generates segmented masks through convolutional operations, including multiplication and concatenation of multi-level features, TCU-Net showed better performance.

Furthermore, we evaluated the case where MDR blocks for refinement of localized spatial information of the transformer features are not included and the case where the inception and attention modules are not included in the middle of the cascaded decoder. The proposed approach with MDR blocks and inception and attention modules verified its superiority over the methods without them. In particular, even if the MDR blocks were applied to the U-Net decoder, much performance improvement was achieved compared to the original U-Net decoder. Another important characteristic of the proposed TCU-Net is that the number of channels of features is reduced, although MDR blocks are added. The number of parameters

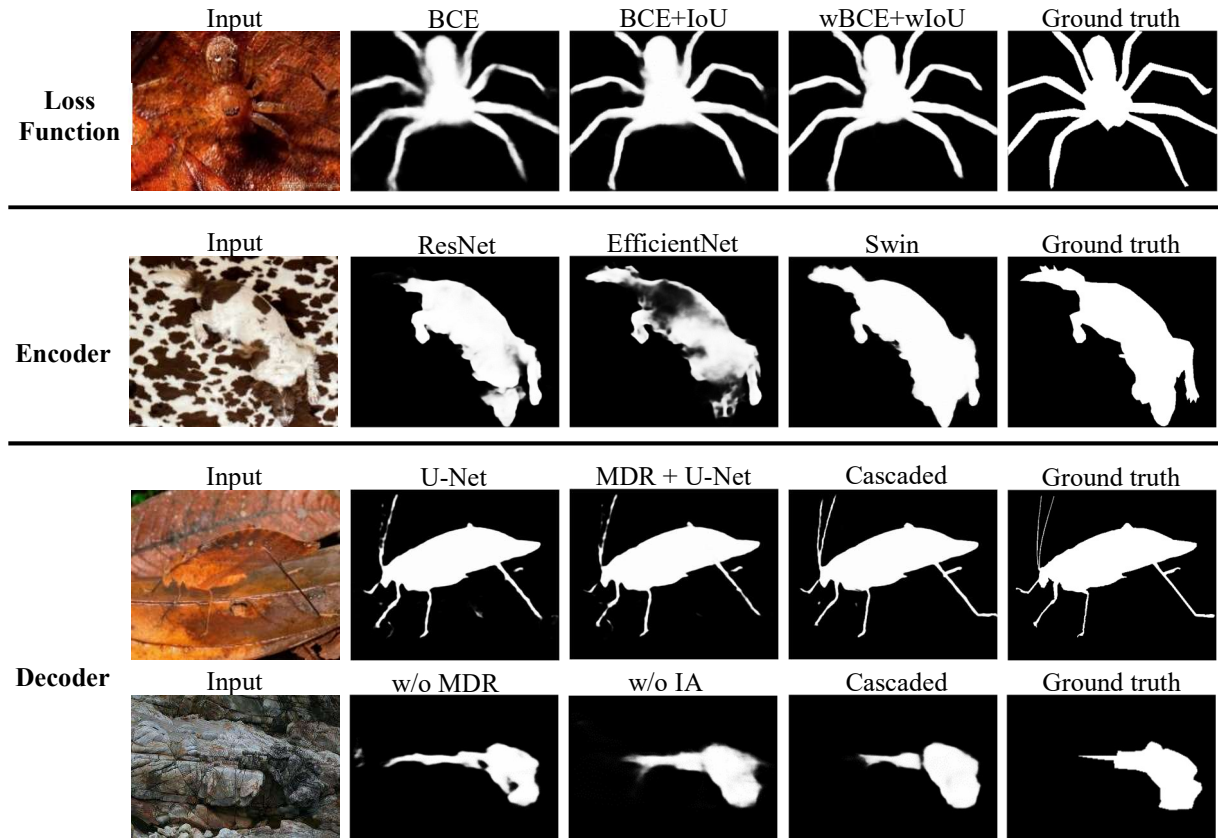


FIGURE 10. Qualitative comparison of the proposed TCU-Net with previous approaches.

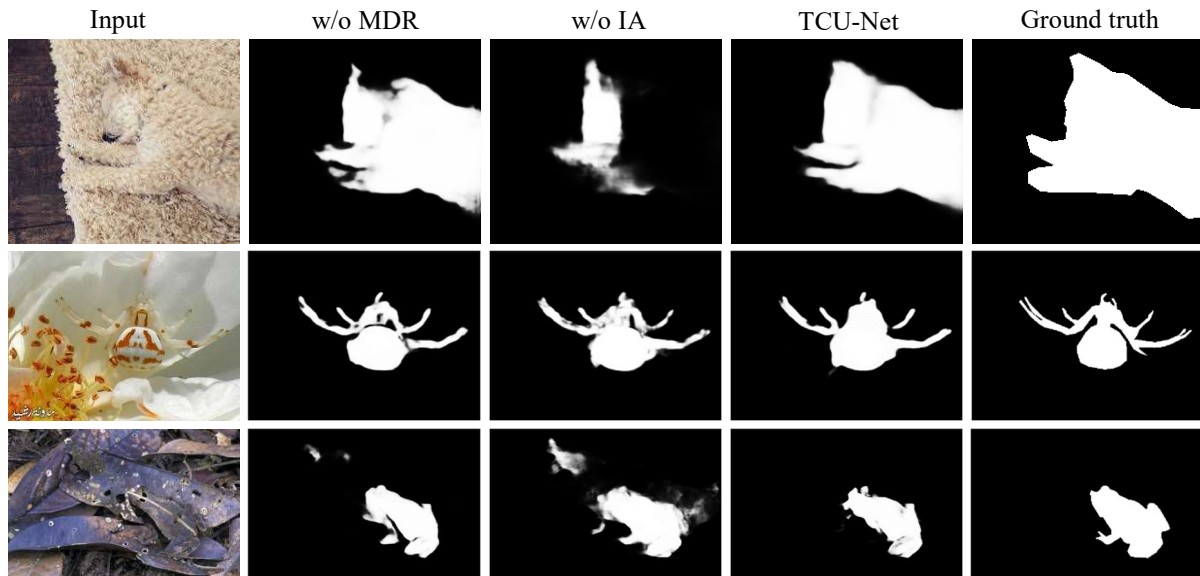


FIGURE 11. Effectiveness of the proposed MDR blocks and IA modules.

without MDR blocks is about 108M, while the number is about 91M with MDR blocks. By using the MDR blocks, not only performance can be improved, but also the memory size can be reduced.

The qualitative comparison of the ablation study is shown in Fig. 10. The first and second rows show the visual comparison of the study concerning loss functions and

encoder architectures. The proposed method (fourth column) generated similar results to the ground truth with the least noise and blur. The third and fourth rows show the results depending on different decoder architectures.

The proposed method could extract thin and fine areas more accurately and generate less noise and blur masks compared with the methods that did not include MDR blocks and

inception and attention modules. As shown in Fig. 11, TCU-Net without MDR blocks generates many unnecessary holes. Furthermore, TCU-Net without the inception and attention modules generates many blur areas. On the other hand, the proposed TCU-Net with MDR blocks and the inception and attentions module can filter them and reduce different types of noise effectively. The ablation study proves the superiority of the proposed approach to previous state-of-the-art studies.

V. CONCLUSION

This study proposed the TCU-Net that effectively converges a Transformer-based encoder and a cascaded CNN-based decoder. Since it is challenging to distinguish the vague boundary between foreground and background, camouflaged object detection is still challenging. The proposed TCU-Net can extract globalized semantic features through the Swin Transformer-based encoder and localized spatial features from the attentive inception decoder with multi-dilated residual (MDR) blocks connecting the encoder and decoder. In particular, the attentive inception decoder consists of the inception module that learns multi-level features and the attention module that performs feature refinement through self-attention. For the loss function, not only weighted BCE loss for learning pixel-wise difference but also weighted IoU loss for learning patch-wise difference are adopted to perform more accurate segmentation. The superiority of the proposed TCU-Net to previous studies was confirmed through comparative evaluation based on two seen and two unseen datasets. The ablation study also verified the effectiveness of the proposed loss function and network architecture.

For future studies, we intend to apply the proposed TCU-Net method to specific fields, such as manufacturing, military, and medical applications. It is still necessary to improve the network for accurately detecting hard-to-find objects, which can be used for supporting domain-specific tasks more intelligently. Furthermore, the lightweight network and reducing calculation time are important topics in the field of deep learning. Further research work is needed to balance the relationship between accuracy and complexity.

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